R13

Code No: 111AD

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year Examinations, October/November - 2020
ENGINEERING PHYSICS
(Common to CE, EEE, ME, ECE, CSE, EIE, IT, MMT, AE, AME, MIE, PTM, AGE)

Time: 2 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Obtain an expression for the packing factor of FCC structure.
b) List various types of bonding in solids with suitable examples.
c) A crystal of BCC structure has atomic radius 1.2 A^0. Find the volume of its unit cell. [4+4+7]

2.a) Explain Braggs law of X-Ray diffraction.
b) Distinguish between Frenkel and Schottky defects.
c) Copper has FCC structure with lattice constant 0.36nm. Calculate the interplanar spacing for (3 2 1) plane. [4+4+7]

3.a) Derive an expression for the de-Broglie wavelength of an electron.
b) Derive time independent Schrödinger's wave equation for a free particle.
c) Calculate the de Broglie wavelength of an electron of energy 100 eV. [4+4+7]

4.a) Explain Bose-Einstein distribution.
b) Discuss the salient features of Kronig-Penny model of a crystal.
c) Explain the concept of effective mass of an electron. [4+4+7]

5.a) Describe different types of polarization mechanisms.
b) What is meant by local field in dielectrics and how it is calculated for a cubic structure?
c) Mention few applications of dielectric materials. [4+4+7]

6.a) How are superconductors classified based on their behavior in the magnetic field?
b) Define magnetic moment. Explain the origin of magnetic moment at the atomic level.
c) Explain Meissner effect. [4+4+7]

7.a) How do you obtain circular rings in the Newton rings experiment?
b) Explain Fraunhofer diffraction due to single slit.
c) Describe the construction and working of Nicol prism. [4+4+7]

8.a) Derive an expression for the density of holes in an intrinsic semiconductor.
b) What is the Hall Effect? Obtain an expression for Hall co-efficient for n-type of semiconductor.
c) Mention the factors affecting the acoustics of buildings. [4+4+7]

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